Online Quiz System — Software Architecture

Layered architecture of a C++17 console application (src/*.h/.cpp, one pair per module; QZ-NF-004)

Player / Administrator • Terminal

keyboard input / text output (stdin, stdout)

PRESENTATION LAYER

«composition root / controller»

main.cpp

Main loop: dispatches menu choices to the modules

Leaderboard (top 10) and player history views

menus

«component»

MenuModule

Main & results menus

Category / difficulty / name

QZ-F-001..003

uses

runQuiz()

summarize()

run()

printHeader()

CORE APPLICATION LAYER

«component»

QuizEngine

Filter & pick 10 questions

Shuffle options per attempt

Check answer / skip, feedback

QZ-F-004, F-010..015

«component»

ScoringModule

Score and percentage

Grade: Excellent / Good /

Needs Improvement

QZ-F-020..022

«component»

AdminModule

Password login: salted, iterated SHA-256

3 failed tries → locked until restart

View / add / edit / delete questions

QZ-F-030..034, SR-001/003/006

uses

Question type

loadQuestions() • saveResult() • loadResults()

load / save questions

CROSS-CUTTING

«utility»

InputValidation

parse*: pure checks

(int, menu choice, A–D/S,

yes/no, free text)

read*: prompt and

re-prompt until valid;

throws InputClosed on EOF

Used by: main.cpp,

MenuModule, QuizEngine,

AdminModule

QZ-F-012, F-042, NF-003

DATA PERSISTENCE LAYER

«component»

FileIOModule

Parse and validate question bank & results; skip malformed lines with warnings

Atomic save: write temp file, then rename • QZ-F-023..025, F-040/041, NF-002

password hash
(direct file I/O)

read / write

append / read

LOCAL STORAGE (data/)

«file»

questions.txt

ID|Category|Difficulty|Question|A|B|C|D|Ans

«file»

results.txt

Player|Category|Score|Total|Timestamp

«file»

admin.cfg

salt + digest (git-ignored)

BUILD & VERIFICATION

CMake (C++17)
quiz_modules static lib
+ quiz executable

ctest suites
unit • component • system

GitHub Actions CI
Linux • macOS • Windows

call / data flow

compile-time type / helper dependency

Calls point downward (except the printHeader() helper); main.cpp wires every module together.